

COM S 127 - Assignment #3 Grading Rubric

Please see the 'Course Schedule' on the Syllabus for this assignment's due date.

Assignment Objective

The purpose of this assignment is to give students practice making a project from a given specification, while slowly building up their code in an iterative manner.

Instructions

Students should study the file(s) provided, and notice the various 'TODO' comments contained therein. These comments indicate places in the file that students should modify the code to be their own. These 'TODO' comments are the places in the script where code/ student work will be evaluated for the final grade on the assignment.

Students' work should be their own, completely original effort.

Files Provided

spinnerWinner.py

Example Output

```
Spinner Winner!
```

```
By: Matthew Holman  
[COM S 127 A]
```

```
Choice? [p]lay, [i]nstructions, [q]uit: p  
Number of Players? [1] / [2]: 1  
#####
```

```
~~ Starting New Round ~~
```

```
** Your target value is: 6
```

```
Press [Enter] To Continue...
```

```
* Player 1 has 10 Points!
```

```
* Player 1 (HUMAN) - How Many Points To Wager?: 3
```

```
Press [Enter] To Continue...
```

```
* Player 2 has 10 Points!
```

```
* Player 2 (AI) - How Many Points To Wager?: 4
```

```
Press [Enter] To Continue...
```

```
* The 'target value' is: 6
```

```

* A single spinner can produces values between 1 - 3
* Player 1 (HUMAN) - How Many Spinners Would You Like To Spin (1 - 3)?: 3

* Player 1 has spun a 2
* Player 1 has spun a 2
* Player 1 has spun a 2
* The total value of the spinner(s) was: 6, and the 'target value' was 6

Press [Enter] To Continue...

* The 'target value' is: 6
* A single spinner can produces values between 1 - 3
* Player 2 (AI) - How Many Spinners Would You Like To Spin (1 - 3)?: 2

* Player 2 has spun a 1
* Player 2 has spun a 3
* The total value of the spinner(s) was: 4, and the 'target value' was 6

Press [Enter] To Continue...

** Player 1 Won! **

Press [Enter] To Continue...

* Player 1 has 13 Points!

* Player 2 has 6 Points!

Press [Enter] To Continue...

~~ End Of Round ~~

```

Special Notes

The 'algorithm' of the game is thus:

- The user selects a game between one or two players.
 - A one player game has the user play against the computer.
 - A two player game has two users play against one another.
- The players start the game with a certain number of 'points.'
- The game is divided into 'rounds.'
 - At the start of the 'round,' the computer generates a 'target value.'
 - This is the value that the players try to match.
 - At the start of the 'round,' each player 'wagers' a certain number of 'points.'
 - Players cannot 'wager' more 'points' than they have. Nor can they 'wager' zero (0) 'points.' Nor can they 'wager' a negative number of 'points.'
 - Then each player decides how many 'spinners' to spin.
 - Each spin adds its value to a final 'spin value' for the round.
 - Players can not spin more spinners than are available to be spun. Nor can the spin zero (0) spinners. Nor can they spin a negative number of spinners.
 - After each player spins their 'spinners,' their 'spin value' is compared against the 'target value.'
 - The player who gets their 'spin value' closest to the 'target value' is the winner of the round.
 - If both players 'spin value' are equally distant from the 'target value' the round is a 'draw.'

- Once the winner is decided, 'points' are added and subtracted from each player's score.
 - The winning player gets their wager amount added to their score.
 - The losing player gets their wager amount subtracted from their score.
- The game continues until one player is completely out of 'points.'

NOTE: While there can certainly be an infinite amount of 'nitpicking' and 'nuance' applied to the algorithm above, the ultimate point is to produce a game that both works correctly, and makes sense.

- The student may have to use their own best judgement when making programming/ design decisions on this project.
- If the student's submission deviates a bit from what is demonstrated in the output above, then that is fine. However, the student's submission should not be radically different.

NOTE: This is not a zero-sum game.

- Meaning, if one player gains a point, this does not necessarily mean that the other player loses a point.
- The points that each player wagers are *not* placed into a single 'pot' that the winner takes at the end of the round. Rather, the players are playing against 'the house,' who has an unlimited number of points to disburse.

NOTE: The `spinnerWinner.py` file provided can be discarded entirely if the student wishes to make their own original version of the game, adhering to the outline above.

- So long as the requisite grading items are completed, then, so long as the submission works correctly, the student should receive full credit for their efforts.
- The output of the student's submission does *not* need to exactly match the output seen above, but it should be readable to the user.
 - A `wait()` function has been included in the `spinnerWinner.py` file to allow the user to 'pause' the game for them to read each line of relevant text.

NOTE: This assignment will be more difficult than the previous assignments.

- Completing this assignment may require the student to start their work 'before the last minute.' Please plan accordingly.
- The student's script **CANNOT** crash under any circumstances. Any portions of the student's code where the script crashes will receive a zero (0) for that aspect of the script.

NOTE: The student is turning in their CODE - NOT the OUTPUT of their code.

- The student's code will be run by the TAs, and the output of those runs, along with the code itself, is what will be evaluated.

NOTE: Assignments turned in in any other format other the specified file types will not be accepted.

- Screenshots of code **will not** be accepted.
- `.sln` files are **not** code files - they contain **no** Python code and **will not** be accepted.
- `.zip`, `.rar`, `.tar.gz`, and other compressed files **will not** be accepted unless otherwise specified.

- If a student's submission is not in a .py file, when so specified, and if the submission is not accompanied by **all** the files needed to run the submission, the submission will not be graded.
 - **THIS WILL LEAD TO THE STUDENT RECEIVING A ZERO (0) ON THE ASSIGNMENT.**
 - **Students will *NOT* be allowed to re-submit their work after the final deadline in this case.**

NOTE: Some students have reported that submitting an assignment on Canvas may have difficulties associated with it.

- Submitting the assignment on Canvas may result in a 'spinning blue circle' on the submission page. This is the normal/ expected behavior.
- Students should submit their work via the GUI built into canvas - not by attempting to access the Canvas API. Doing so may cause problems.
- So long as the top of the page contains text that reads 'Submitted,' there should not be a problem.
- The student can confirm the status of their submission by downloading it themselves *after* they submit it. This is **highly** recommended.
 - **IF STUDENTS SUBMIT THE INCORRECT FILE(S), THEY CAN ALWAYS RESUBMIT *BEFORE* THE FINAL DEADLINE. STUDENTS WILL *NOT* BE ABLE TO RESUBMIT *AFTER* THE FINAL DEADLINE.**
- Students may appear to encounter difficulties in submitting their assignment due to issues with the Canvas platform itself. However, this will **not** be an excuse to turn in work past the deadline.
 - **THIS MEANS THAT STUDENTS SHOULD NOT WAIT UNTIL THE 'LAST MINUTE' TO ATTEMPT TO SUBMIT THEIR ASSIGNMENT TO CANVAS. THUS, STUDENTS WILL NEED TO 'PLAN THEIR LIVES' IN REGARD TO THEIR TIMELY SUBMISSION.**
 - **STUDENTS SHOULD ATTEMPT TO SUBMIT THEIR WORK *BEFORE* THE FINAL DEADLINE.**
 - **IF STUDENTS ENCOUNTER PROBLEMS SUBMITTING THEIR ASSIGNMENTS, PERHAPS TRY A DIFFERENT BROWSER, CLEAR THE BROWSER CACHE, ETC.**
 - **IF SUDENTS NEED HELP TURNING IN THEIR ASSIGNMENTS, THEY ARE ALWAYS WELCOME AT INSTRUCTOR/ TA OFFICE HOURS.**
 - **MANY ASSIGNMENTS HAVE BEEN SUBMITTED IN THE PAST, THUS IT *IS*, INDEED, POSSIBLE TO DO.**

Grading Items

- **(Deadline)** Was the assignment turned in by the 11:59 p.m. Friday deadline and was the file named `spinnerWinner.py`? _____ / 1
- **(Name at Top)** Has the student typed in their name/ date at the top of the `spinnerWinner.py` file and deleted this 'TODO' comment?: _____ / 1
- **(Assignment #/ Description)** Did the student write the assignment number and a brief description of what the assignment is inside the initial comment block, and then delete this 'TODO' comment?: _____ / 1
- **(Name/ Section Number)** Has the student added their name/ section number to the initial script output?: _____ / 1
- **('p' option tasks)** Has the student completed the work items specified by the 'p' option tasks section?: _____ / 4
- **('i' option tasks)** Has the student completed the work items specified by the 'i' option tasks

section?: _____ / 1

- (**'q' option tasks**) Has the student completed the work items specified by the 'q' option tasks
section?: _____ / 1

TOTAL _____ / **10**